Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/10/2012
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Project Title:	Molecular and Cellular Mechanisms of Space Radiation-Induced Changes in Hippocampal-Dependent Behavior		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Previously at University of Texas Southwestern Medical Center at Dallas, 2000-2016		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	11/10/2011	**End Date:**	11/09/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NNX12AB55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

An unavoidable consequence of many space missions is exposure to galactic cosmic radiation (GCR), which includes HZE particles like Fe, Si, and O. Estimating radiation risks to the central nervous system (CNS) is a high research priority. Here we propose extensive analysis of how a specific region of the CNS, the hippocampus, is influenced by radiation. The hippocampus is important because it contributes to spatial learning and mood regulation, which are critical for mission success. Recently the hippocampal dentate gyrus (DG) was shown to be involved in learning challenging tasks requiring fine spatial distinction ("pattern separation"). The hippocampus is also important because it is exceptionally "neuroplastic". For example, in terms of gene transcription changes, the DG is among the most responsive CNS regions to physiological and environmental stimuli. The DG is also a site of neurogenesis throughout life, and adult neurogenesis is proposed to contribute to hippocampal function in general and pattern separation in particular. Understanding how space radiation influences DG structure and function will allow far better estimation of CNS radiation risks. Research already suggests that space radiation has a negative effect on hippocampal structure and function. However, it is unknown whether space radiation specifically influences DG functions like pattern separation. It is also unknown whether the diverse neural cells types within the DG are similarly influenced by space radiation or whether, for example, there is a cell-type specific change in translation or morphology. We will use recent advances in knowledge of DG function and structure and cutting-edge but proven technical approaches to answer key remaining questions about how space radiation influences the cellular, molecular, and functional integrity of the hippocampus.
For example in Aim 1, using behavioral tests with which we have recently published (delayed non-match to pattern radial arm maze for fine spatial distribution; Guo et al., Nat Med 2011), we will explore our hypothesis that space radiation decreases pattern separation in a dose- and LET-dependent manner.

In Aim 2, using transgenic mice we have developed that allow inducible YFP labeling of neural stem cells and their progeny (nestin-CreERT2/R26R mice; Lagace et al., J Neurosci 2007), we will explore our hypothesis that space radiation negatively impacts both adult-generated (Aim 2a) and embryonic-generated (Aim 2b) DG neurons.

In Aim 3, using transgenic mice developed by our colleagues that allow profiling of translating RNA in genetically-defined DG neural cells, we will explore our hypothesis that space radiation causes discrete changes in DG neural cell types in a dose- and LET-dependent manner. Unlike currently used approaches to assess total RNA profiles from cellularly heterogeneous samples (e.g. laser capture dissection and subsequent microarray analysis), this approach allows purification of specific cell types via ribosome tag and a quantitative comparison of translating RNA profiles in genetically-distinct DG cells.

Importantly, all mice will be 6-9 month old at time of radiation exposure in order to better assess the impact of ground-based space radiation on the fully mature brain. Tissue from Aims 1-3 will be examined using immunohistochemistry, epifluorescent and confocal microscopy, stereologic cell counting, and morphological analysis, and tissue from Aim 3 will be explored using microarrays on the translating RNA pools and qRT-PCR.

Taken together, these aims will allow us to test our overall hypothesis that radiation will have distinct, long-lasting effects on the behavioral, cellular, and molecular integrity of the hippocampus. Thus, these results will expedite the quest to estimate CNS radiation risks due to GCR exposure.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	It is critical to identify factors that regulate normal brain function and structure. Our proposed research will elucidate how ground-based space radiation influences hippocampal structure and function. The knowledge derived from the proposed experiments will provide insight into the mechanisms contributing to brain and hippocampal function and their disruption by space radiation, and thus has the potential to shed light on how we can estimate CNS risk from space radiation. This work is important given the long-duration space missions on the former Mir Space Station, the present International Space Station, and the potential exploratory missions to the Moon and Mars. Mice will be used as the primary source of nervous tissue for experimentation since most of the relevant literature concerns studies of tissues of rodent origin, and since the mouse is an excellent model for human exposure to space radiation given their remarkably similar brain structure and function and genomic similarity. At this time, using computer modeling to understand the effects of space radiation on behavioral, cellular, and molecular indices of the hippocampus is just not possible given the complexity of the CNS. Cell culture systems have proven extremely useful to ask mechanistic questions about the effects of space radiation on the hippocampus, but they are unable to address the effects of space radiation on the whole organism, particularly behavior. The combination of approaches used here - analysis of behavior, cellular morphology, and translational profile - will have significant benefits for the fields of neuroscience in general and the fields of learning and memory, stem cell biology, and even neural development in particular. The particular focus of this application - hippocampal dysfunction - is evident in a range of psychiatric and neurologic disorders, including Alzheimer's Disease, depression, and addiction. Astronauts may be particularly at risk for delayed hippocampal pathological signs, as it is estimated that even just 1 year in a low-Earth orbit mission may result in 91% of hippocampal cells and 25% of hippocampal cell-nuclei hit by high LET particles. These studies will help us predict neurodegenerative or cognitive changes in astronauts after mission completion. We also focus on the hippocampus since it is exceptionally "neuroplastic" – being among the most responsive brain regions in terms of gene transcription after physiological manipulation and stimuli and being a site of the birth of new neurons throughout life. This exceptional neuroplasticity suggests that the hippocampus may be among the first brain regions to reveal a response to radiation. To best study the influence of ground-based space radiation on hippocampal structure and function in a way that is most translational from rodents to humans, we will be specifically using older mice (e.g. ~6-9months old at time of irradiation). Aside from the general benefits of this broader work for neuroscience and humans, there are several innovative aspects to this work. For example, using mice that we have generated that allow inducible labeling and tracking of embryonic vs. adult-generated neurons, we will be able to for the first time tease apart effects on these two populations. In addition, using novel mice that allow profiling of translating RNA in genetically-defined DG neural cells (e.g. adult-generated vs. embryonic-generated DG neurons), we will make a quantitative comparison of translating RNA profiles in genetically-distinct DG neural cells in a time-, dose-, and LET-dependent manner. Utilization of these novel technologies have the implication to expand our broader understanding of mechanisms underlying neuroplasticity and even neurodegenerative processes.
Task Progress:	During the first 10 months of this new grant, we were able to make good progress on our aims and lay the critical foundation for each of the three aims. For Aim 1 (how does ground-based space radiation influence dentate gyrus and hippocampal function?), we have examined both control and irradiated mice in two different hippocampal-dependent tasks, delayed non-match to place radial arm maze (DNMP-RAM) and contextual fear conditioning (FC). While the subject numbers are too small to conclude anything yet (e.g. less than 10 per group), these studies have allowed us to finetune the protocols for the behavioral tests. We are now prepared to test larger groups in these hippocampal-dependent tasks. Also for Aim 1, we evaluated effect of ground-based space radiation on key physiological parameters (body mass, locomotor activity, open-field behavior). First we assessed animals exposed to 28Si irradiation (300MeV/n). Consistent with our expectations, we found no effect of irradiation on these basic measures when examined at ~9months of age. Next we assessed animals exposed to 56Fe (600MeV/n, up to 100cGy). These analysis are currently being performed (September 2012). For Aim 2, (how does radiation influence the dendritic morphology of embryonic- vs. adult-generated granule cell neurons?), we have utilized our novel nestinCreERT2/R26R-YFP mice to begin to finetune our protocol to differentially track and label these two different populations of cells. Pilot mice will be sent for radiation exposure in Fall, 2012. For Aim 3, (what is the impact of radiation on the translational profile of genetically-defined dentate gyrus neural cells?), we have received and begun breeding several lines of the transgenic mice needed for this study. We anticipate having sufficient numbers to send the first cohort of mice in 2013 for radiation exposure. We are also making excellent progress on publications, as we already have one publication under revision (In vivo contribution of nestin- and GLAST-lineage cells to adult hippocampal neurogenesis) and several more in preparation. This is due in part to the prior grant shared between Dr. Eisch and her co-I on this grant, Dr. Benjamin Chen, who was PI on NNA07AP84G, which has given the research team significant experience with these experiments. We have also presented oral/poster presentations at several conferences, including: The Annual Meeting of the Society for Neuroscience (November, 2011) ; The NASA Space Radiation ad hoc Expert Panel on CNS Risks (January, 2012) ; The 23rd annual NASA Space Radiation Investigator Workshop (July, 2012).
Bibliography Type:	Description: (Last Updated: 11/17/2025)
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